

INTEGRATED ELECTRICAL SYSTEMS OF SOLAR PHOTOVOLTAIC TECHNOLOGY

LABORATORY 3

GROUP 2



TÉCNICO LISBOA

**The Impact of Non-Uniform Irradiance at Photovoltaic Arrays and the Role of Bypass
Diodes**

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1 Introduction

The primary objective of this laboratory experiment is to investigate the impact of partial shading and irradiance variations on the current-voltage and power-voltage characteristics of photovoltaic cells and arrays. To achieve this, the experimental procedure involves reducing the incident light in one cell by covering it with a paper at different levels. Because the cells are connected in series, the exact same current must flow through the entire circuit. Consequently, when one cell produces less current than the other, it limits the overall output of the array. If the current produced by the unshaded cell exceeds the capacity of the shaded cell, it forces the shaded cell to conduct the excess current, driving it into reverse bias. Under this condition, instead of generating power, the cell acts as a load, dissipating the excess energy as heat and creating a hot spot. To mitigate this issue, a bypass diode can be connected in parallel in each cell. Under uniform irradiance, the cells operate in forward bias while the bypass diodes remain in reverse bias. However, when a shaded cell is forced into reverse bias, the negative potential difference across it activates the bypass diode. This provides an alternate path for the current, however, the voltage of the shaded cell no longer contributes to the array's total voltage.

Before proceeding with the laboratory experiment, a LTSpice simulation was performed to establish a theoretical reference for the expected results. Two 1D5P equivalent circuits were connected in series, representing two different cells, Figure 1.

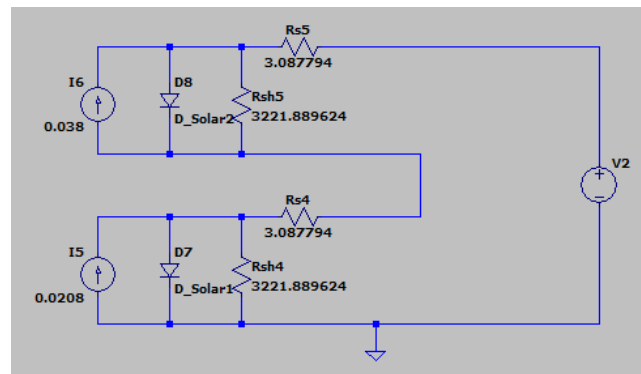


Figure 1: Two cells in series

In order to simulate partial shading, different values of I_{ph} were applied to the two cells. Two scenarios were simulated: the first one, Figure 2, with no shading and the second with 50% shading on the first cell, Figure 3.

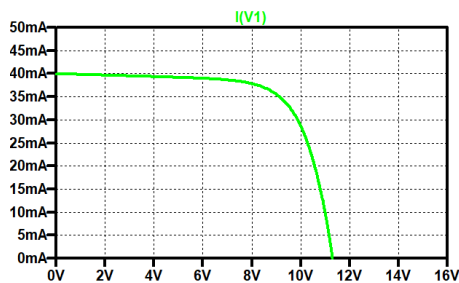


Figure 2: No shading

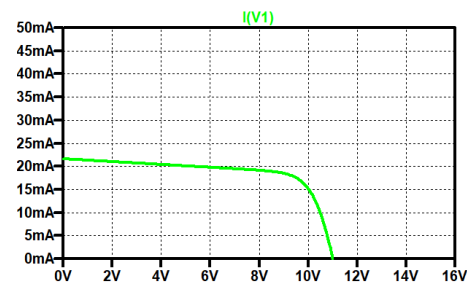


Figure 3: 50% shading in first cell

It is possible to visualize that when a cell is shaded, the current produced by that cell is reduced, so the overall circuit current is limited by the shaded cell.

This effect would lead into hot spots in the shaded cell, degrading it. However, this issue can be mitigated by applying a bypass diode in parallel with each cell, Figure 4.

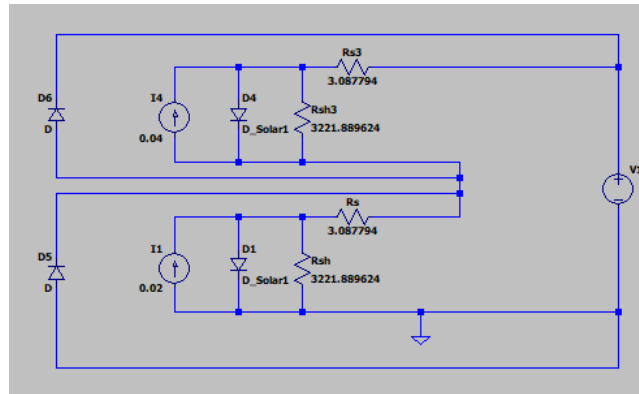


Figure 4: Two cells in series with bypass diodes

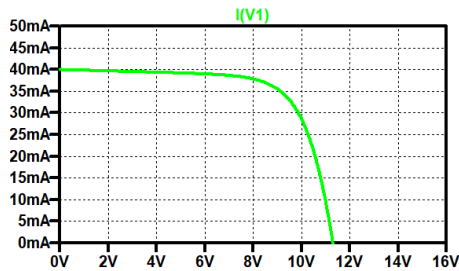


Figure 5: No shading

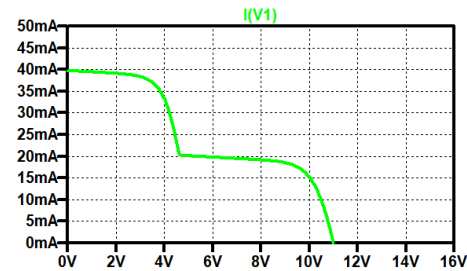


Figure 6: 50% shading in first cell

By introducing both bypass diodes, the behavior of the system is completely different. In the unshaded scenario, the I-V curve remains the same, as their bypass diodes remain inactive. However, in the shaded scenario, the resulting I-V curve consists in two different current values.

At lower voltages, the unshaded cell forces its current to flow in the circuit. This drives the shaded cell into reverse bias, activating the bypass diode. The string maintains a high current output but the shaded cell's voltage isn't contributing to the total array's voltage. As the operating voltage increases and surpasses the capacity of the unshaded cell, the system demands voltage contribution from the shaded cell. Consequently, the current drops to the shaded cell's current value, creating the step in Figure 6.

Consequently, the stepped I-V characteristic changes the Power-Voltage (P-V) curve. While an unshaded array has only one power peak, Figure 7, the shaded array with bypass diodes develops multiple maximum points, Figure 8. The highest value on this curve is the Global Peak, representing the optimal operating condition. Any lower peak is considered a Local Peak. This visually confirms that the implementation of the bypass diode recovers an extra portion of the power that otherwise would be lost.

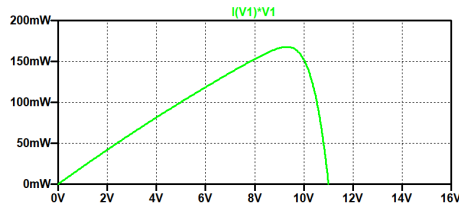


Figure 7: P-V curve with no shading

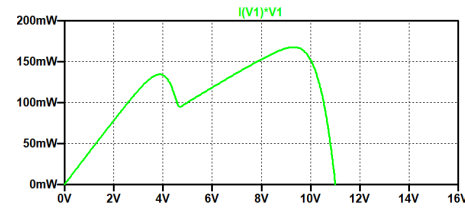


Figure 8: P-V curve with 50% shading in the first cell

2 Results

All tests were conducted using a projector as the light source, placed at 35 cm from the table and the measured irradiance was 1015 W/m^2 , trying to match the irradiance from the previous lab (1020 W/m^2).

The experimental setup used in this laboratory was similar to the one described in the first experiment. The I-V characteristics of the array were captured using the same load sweep methodology. By manually adjusting a variable resistor, connected in series, the operating point of the array was transitioned across the curve.

In order to perform the I(V) curve, MATLAB was used to fit a smooth trend line into the points of the oscilloscope.

2.1 I(V) and P(V) Characteristic of 2 Photovoltaic Modules Connected in Series

Firstly, two modules connected in series were tested. The obtained I-V and P-V curves are the following.

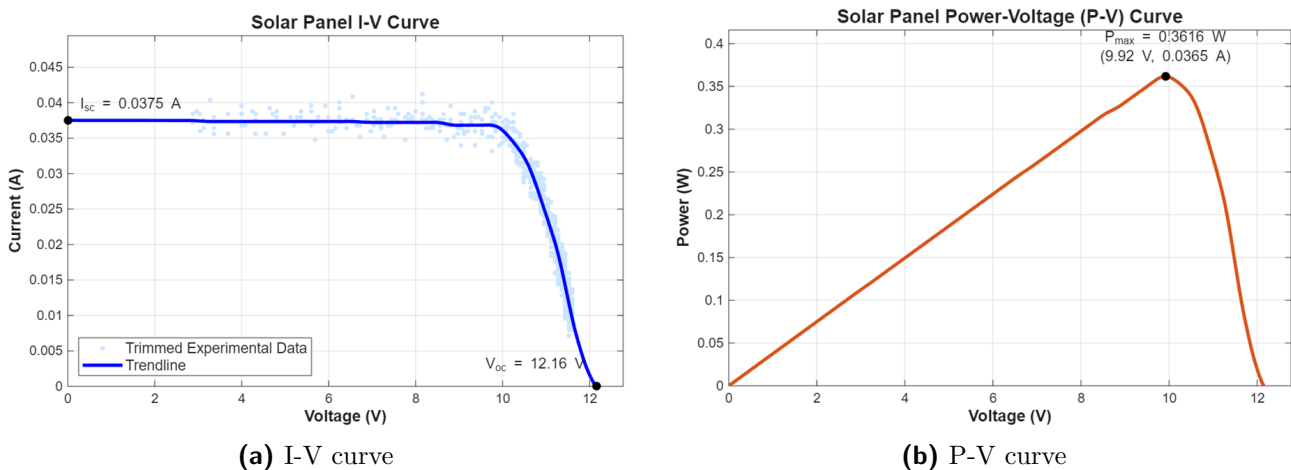


Figure 9: I-V and P-V characteristics of two modules in series

Table 1: Electrical parameters for two modules connected in series (no shading)

Parameter	Value
V_{oc} (V)	12.1619
I_{sc} (A)	0.0375
P_{max} (W)	0.3616
V_{mp} (V)	9.9196
I_{mp} (A)	0.0365
Fill Factor (%)	79.29
Efficiency (%)	4.63

These parameters align with the parameters obtained in the first laboratory experiment, with the exception of the I_{sc} and P_{max} values. This discrepancy could be attributed to a calibration error in the solar meter.

The obtained values are consistent with the results from Laboratory 1 for two modules connected in series. As expected, the open-circuit voltage is approximately doubled when compared with a single module, while the current remains close to the value of one module. The maximum power also increases because the operating voltage at the maximum power point is higher. The fill factor of 79.29% indicates a good quality I(V) curve, with relatively low losses under uniform irradiance.

2.2 I(V) and P(V) Characteristic of 2 photovoltaic Modules Connected in Series, considering 1 Partial Shaded Module

The main objective was to investigate and quantify the impact of different partial shading levels on a single cell within the array. Measurements were taken for the following shade levels: 0%, 25%, 50%, 75% and 100%. The I-V and P-V characteristic curves, captured using the load sweep methodology described previously, are presented below.

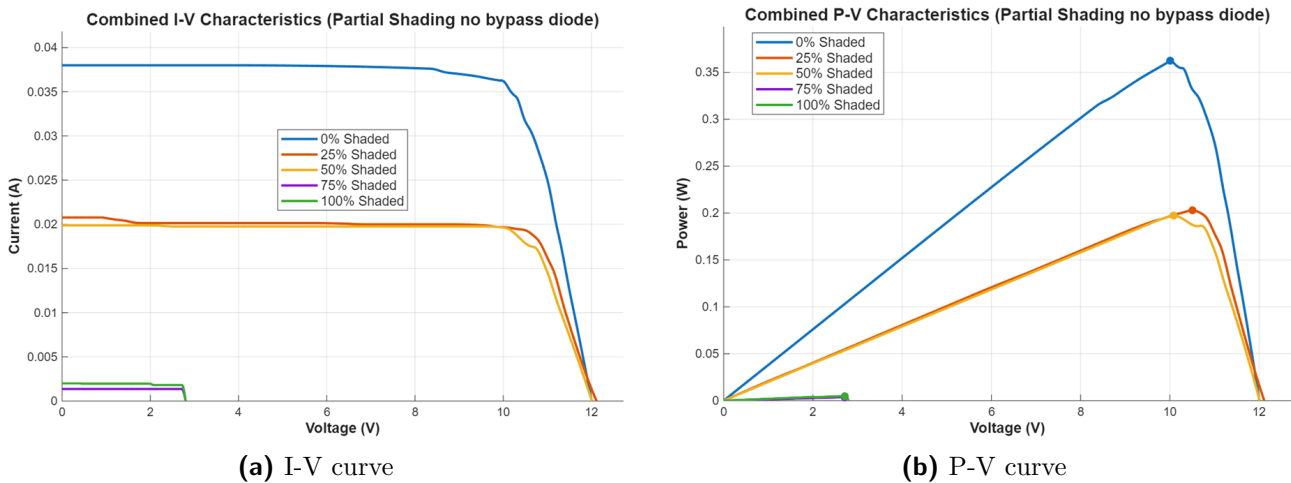
**Figure 10:** I-V and P-V characteristics of two modules in series for different shadings in one module

Table 2: Electrical parameters for two modules connected in series with partial shading on one module (without bypass diodes)

Setup	V_{oc} (V)	I_{sc} (A)	P_{max} (W)	V_{mp} (V)	I_{mp} (A)	FF (%)	Eff (%)
0% Shaded	12.0066	0.0380	0.3622	10.0096	0.0362	79.39	4.63
25% Shaded	12.1143	0.0208	0.2029	10.5120	0.0193	80.65	2.60
50% Shaded	12.0127	0.0199	0.1971	10.0868	0.0195	82.50	2.52
75% Shaded	2.8000	0.0014	0.0037	2.7158	0.0014	96.99	0.05
100% Shaded	2.8000	0.0020	0.0049	2.7158	0.0018	87.71	0.06

By analysing the experimental data, it is possible to verify that the current produced is directly affected by the irradiance. Since the modules are connected in series, the shaded module limits the current of the entire string. This explains the reduction of I_{sc} and P_{max} when the shading level increases.

It is also possible to verify that when the module is 25% or 50% shaded it produces nearly the same amount of current. This can be explained by the fact that, in both cases, the shaded module is already acting as the current-limiting element of the series connection, so the measured current becomes very similar.

For 75% and 100% shading, the power decreases drastically. In these cases, the shaded module can no longer contribute normally to the operation of the string, which leads to very low current and power values. This is the main problem under analysis in this laboratory: under non-uniform irradiance, a shaded module strongly limits the performance of the series-connected array.

**Figure 11:** Efficiency at different load values

This figure shows the evolution of efficiency for different values of resistance. It can be noted that the maximum efficiency corresponds to the maximum power point, with efficiency decreasing as the operation point deviates from this optimum point.

2.3 I–V and P–V Characteristics of Two Series-Connected PV Modules with Bypass Diodes under Partial Shading

For this experiment, two modules were connected in series with bypass diodes. Different shading percentages were tested in one of the modules. The results are displayed in the figures below.

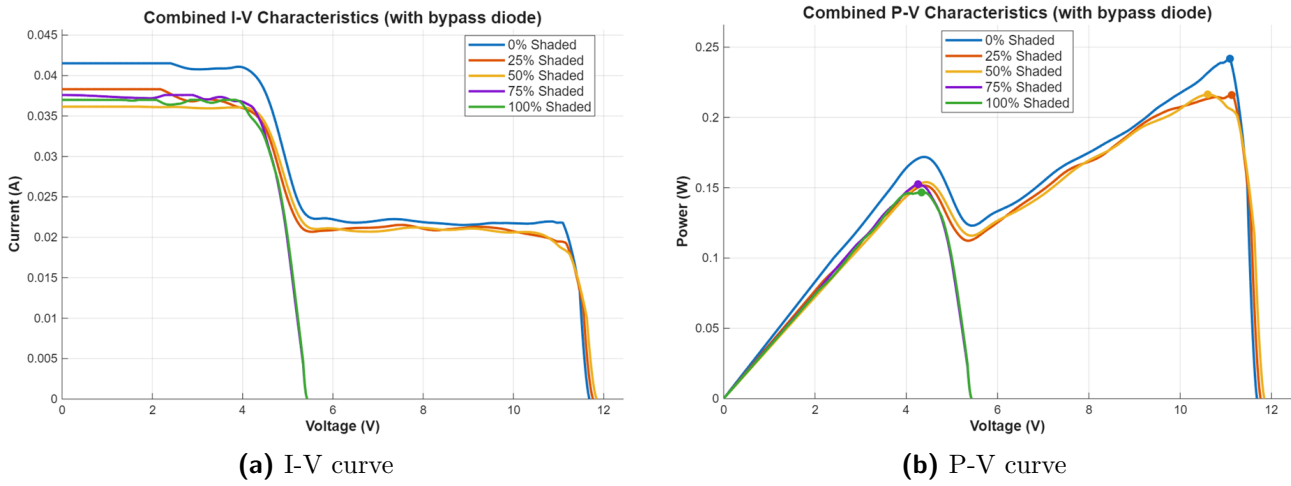


Figure 12: I–V and P–V characteristics of two modules in series for different shadings in one module with bypass diodes

Table 3: Electrical parameters for two modules connected in series with partial shading on one module (with bypass diode)

Setup	V_{oc} (V)	I_{sc} (A)	P_{max} (W)	V_{mp} (V)	I_{mp} (A)	FF (%)	Eff (%)
0% Shaded	11.6800	0.0415	0.2416	11.0948	0.0218	49.83	3.09
25% Shaded	11.7600	0.0383	0.2158	11.1237	0.0194	47.89	2.76
50% Shaded	11.8400	0.0362	0.2163	10.6062	0.0204	50.52	2.77
75% Shaded	5.4400	0.0376	0.1524	4.2517	0.0359	74.53	1.95
100% Shaded	5.4400	0.0370	0.1468	4.3389	0.0338	72.92	1.88

With the introduction of the bypass diodes, the behavior under shading changes significantly. For low shading levels, the results are still close to the unshaded case. However, for 75% and 100% shading, the current does not collapse as in the previous case.

This happens because the bypass diode provides an alternative path for the current when the shaded module cannot sustain the current imposed by the string. As a result, the shaded module no longer contributes significantly to the total voltage, which explains the reduction of V_{oc} to about 5.44 V for the highest shading levels.

Comparing with the case without bypass diodes, the benefit is clear under severe shading. The maximum power for 75% shading increases from 0.0037 W to 0.1524 W, and for 100% shading from 0.0049 W to 0.1468 W. Therefore, the bypass diode reduces mismatch losses and helps prevent reverse bias operation of the shaded module.

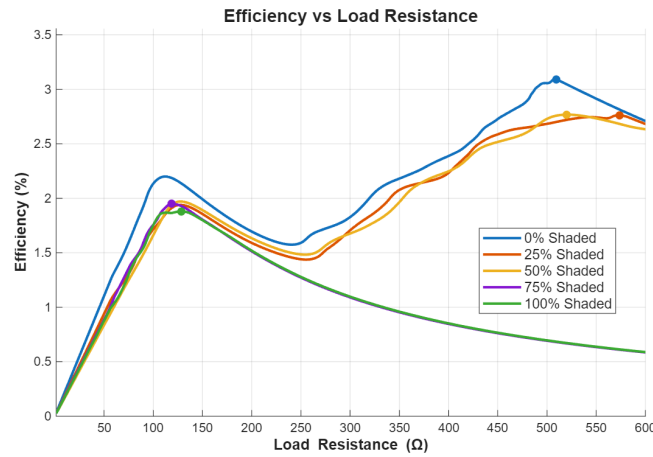


Figure 13: Efficiency at different load values

This figure shows the evolution of efficiency for different values of resistance. It is noted that there are two peaks of efficiency, corresponding to the two steps of the I-V curve due to the use of the bypass diode.

2.4 I–V and P–V Characteristics of Two Series-Connected Photovoltaic Modules with Bypass Diodes under Different Shading Combinations

For this experiment, two modules were connected in series with bypass diodes. Different combinations of shading were tested in the different solar modules. The results are displayed in tables and V_{oc} , I_{sc} , P_{max} and efficiency were evaluated.

Table 4: Open-Circuit Voltage (V_{oc}) for different shading combinations

P1 \ P2	0%	25%	50%	75%	100%
0%	11.6800	11.7600	11.8400	5.4400	5.4400
25%	11.7600	11.6800	11.4400	5.3600	5.5200
50%	11.8400	11.4400	11.2800	5.2800	5.3600
75%	5.4400	5.3600	5.2800	2.4800	2.5600
100%	5.4400	5.5200	5.3600	2.5600	NaN

Table 5: Short-Circuit Current (I_{sc}) for different shading combinations

P1 \ P2	0%	25%	50%	75%	100%
0%	0.0415	0.0383	0.0362	0.0376	0.0370
25%	0.0383	0.0206	0.0213	0.0192	0.0191
50%	0.0362	0.0213	0.0212	0.0200	0.0194
75%	0.0376	0.0192	0.0200	0.0020	0.0020
100%	0.0370	0.0191	0.0194	0.0020	NaN

Table 6: Maximum Power (P_{max}) for different shading combinations

P1 \ P2	0%	25%	50%	75%	100%
0%	0.2416	0.2158	0.2163	0.1524	0.1468
25%	0.2158	0.2000	0.1906	0.0825	0.0845
50%	0.2163	0.1906	0.1855	0.0796	0.0776
75%	0.1524	0.0825	0.0796	0.0043	0.0046
100%	0.1468	0.0845	0.0776	0.0046	NaN

Table 7: Efficiency (η) for different shading combinations

P1 \ P2	0%	25%	50%	75%	100%
0%	3.09	2.76	2.77	1.95	1.88
25%	2.76	2.56	2.44	1.06	1.08
50%	2.77	2.44	2.37	1.02	0.99
75%	1.95	1.06	1.02	0.06	0.06
100%	1.88	1.08	0.99	0.06	NaN

Table 8: Fill Factor (FF) for different shading combinations

P1 \ P2	0%	25%	50%	75%	100%
0%	49.83	47.89	50.52	74.53	72.92
25%	47.89	82.94	78.05	80.31	80.14
50%	50.52	78.05	77.72	75.41	74.64
75%	74.53	80.31	75.41	87.58	89.18
100%	72.92	80.14	74.64	89.18	NaN

When both modules are partially shaded, the electrical parameters show a progressive reduction in performance as the shading level increases. The highest power is obtained for the unshaded case, with $P_{max} = 0.2416$ W, and the lowest values occur when both modules are strongly shaded, reaching only 0.0043 W for the 75%/75% case and 0.0046 W for the 75%/100% case.

Comparing with the previous section, the role of the bypass diode is still evident, since it helps reduce mismatch losses and prevents severe reverse bias operation. However, when both modules are heavily shaded, its effect becomes limited, because there is no sufficiently illuminated module left to sustain significant power generation. Therefore, bypass diodes improve the behaviour under partial shading, but they cannot fully compensate for severe shading affecting both modules.

2.5 I–V and P–V Characteristics of a 2×2 Series–Parallel Photovoltaic Array with One Partially Shaded Module

Series–Parallel Connection

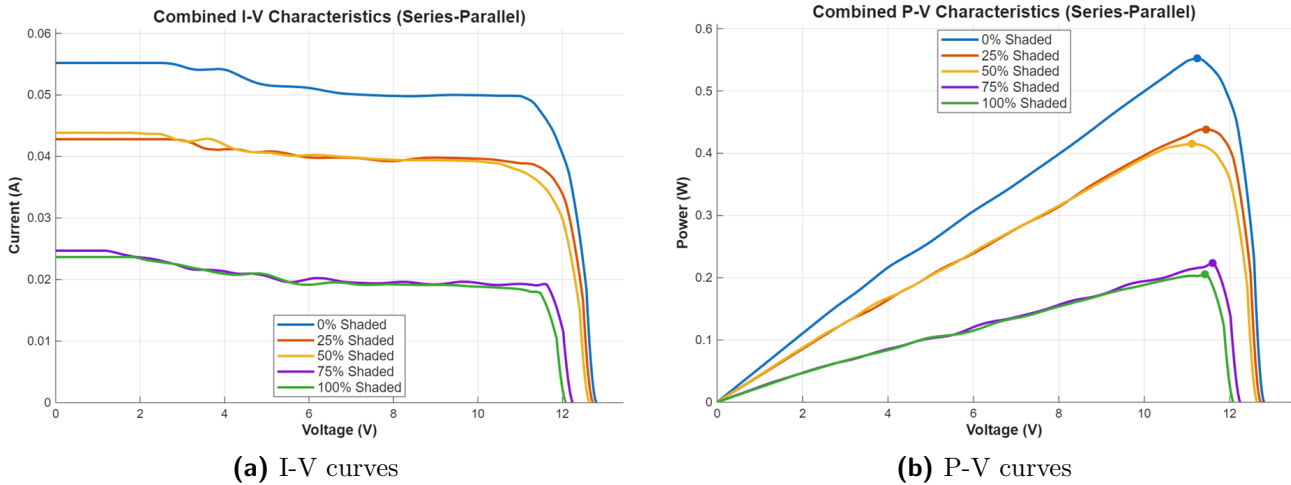


Figure 14: I–V and P–V characteristics of 4 modules in series-parallel with different partial shading percentages on one module

In the series-parallel configuration, partial shading on one module reduces mainly the current capability of the affected branch. As a result, the total array current decreases with increasing shading, while the voltage remains relatively high because the second series branch is still able to contribute to the output voltage. This is reflected in the gradual decrease of I_{sc} from 0.0552 A in the unshaded condition to 0.0237 A at 100% shading, while V_{oc} remains close to 12 V. The maximum power decreases from 0.5521 W to 0.2051 W, showing that the series-parallel configuration is more tolerant to shading than a simple two-module series string, but still suffers significant mismatch losses.

TCT Connection

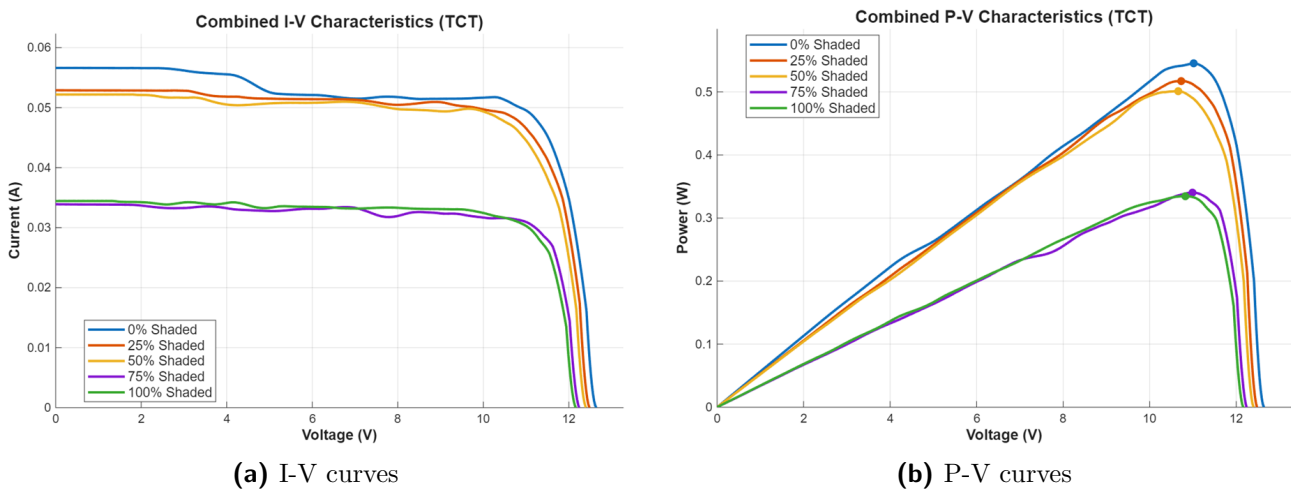


Figure 15: I–V and P–V characteristics of 4 modules in TCT with different partial shading percentages on one module

In the TCT configuration, the effect of partial shading is mitigated more effectively because the additional cross-ties allow a better redistribution of current among parallel paths. This increased redundancy reduces the current limitation imposed by the shaded module and improves the overall power extraction. Even under 100% shading of one module, the array still delivers 0.3346 W, which is substantially higher than in the series-parallel case. The same trend is observed for 25%, 50% and 75% shading, confirming that TCT offers superior resilience under non-uniform irradiance.

Series-Parallel vs TCT

Table 9: Comparison of electrical parameters between Series-Parallel and TCT configurations

Shading	Config	V_{oc} (V)	I_{sc} (A)	P_{max} (W)	V_{mp} (V)	I_{mp} (A)	FF (%)	Eff (%)
0%	S-P	12.80	0.0552	0.5521	11.2353	0.0491	78.14	3.53
	TCT	12.64	0.0566	0.5450	11.0188	0.0495	76.14	3.49
25%	S-P	12.72	0.0428	0.4385	11.4455	0.0383	80.54	2.81
	TCT	12.48	0.0529	0.5176	10.7293	0.0482	78.41	3.31
50%	S-P	12.64	0.0439	0.4152	11.1202	0.0373	74.89	2.66
	TCT	12.40	0.0522	0.5015	10.6605	0.0470	77.49	3.21
75%	S-P	12.24	0.0247	0.2234	11.6022	0.0193	73.89	1.43
	TCT	12.24	0.0339	0.3404	10.9890	0.0310	82.05	2.18
100%	S-P	12.08	0.0237	0.2051	11.4264	0.0180	71.77	1.31
	TCT	12.16	0.0344	0.3346	10.8197	0.0309	79.89	2.14

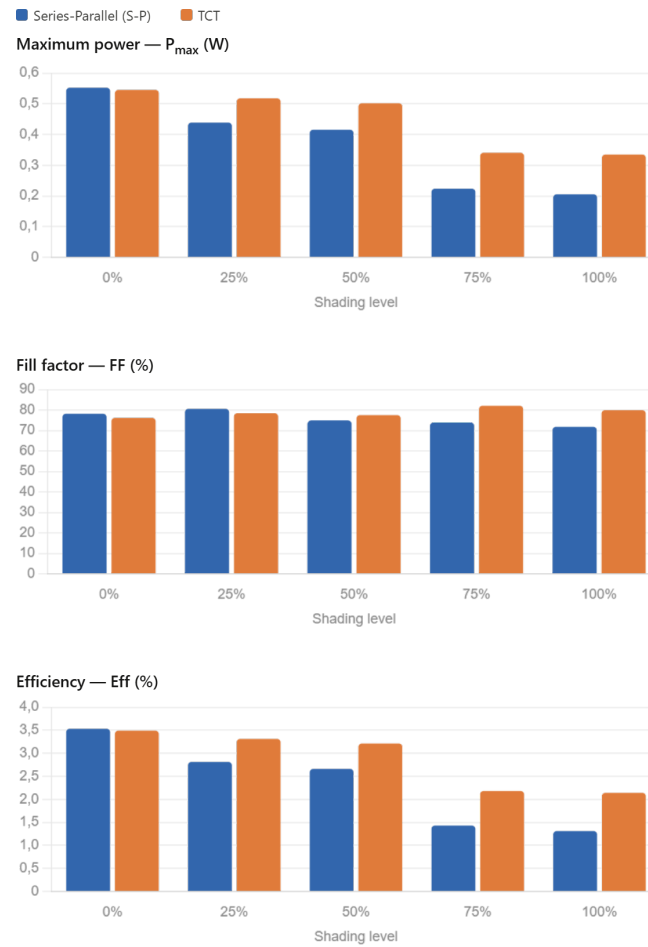


Figure 16: Performance comparison of Series-Parallel and TCT configurations under partial shading

Under uniform irradiance, both configurations present similar results. However, when partial shading is introduced, the TCT configuration performs better in all cases. For example, at 100% shading, the maximum power is 0.2051 W for Series-Parallel and 0.3346 W for TCT.

This difference is mainly due to the higher current maintained by the TCT configuration under shading. Therefore, TCT is the most suitable configuration among the ones tested for mitigating shading effects.

3 Conclusion

This laboratory aimed to evaluate the effect that different percentages of shading does to the performance of the photovoltaic modules. The use of bypass diodes was tested when two photovoltaic modules were connected in series. After that the performance of the 2x2 Series-Parallel connection was tested in comparison with the TCT connection, when one of the solar modules were partially shaded.

For two modules connected in series the use of bypass diodes is crucial to maintain a good power output when shading values are high. It is important to see for example the case where one of the modules was 75% shaded. Without bypass diodes, the power output was 0.0037 W, comparing to a

much higher 0.1524 W when bypass diodes were introduced in the setup.

Comparing the 2x2 Series-Parallel and TCT connections it is evident that TCT is superior as it was able to deliver a much higher power output. For values of 25% and 50% of shading, TCT delivered 20% more power, with this difference reaching 55% at 75% and 100% of shading.